Seaborn Vs MatplotLib Transcript

Hello everyone, welcome to BITS PLANET DIGITAL. Today we will continue our learning journey for the course Data Visualisation Storytelling. We will be now looking at advanced features and other libraries in Python environment for developing even more customised and well-looking graphics.

We will be now looking at the Seaborn library for data visualisation which is there in the Python libraries and try to understand how it can be used over and above the Mathplotlib library for developing better visualisations. So let us get into today's lesson. Like in last few classes, we will be demoing the code directly in the Colab environment and once we share the files with you, you can also practise them and learn from the examples given there for developing visualisations.

Again, the best way to learn the libraries is to practise extensively using different examples for which you can go to the source documentation of Seaborn library and practise from there. Now let us start with understanding the difference between Mathplotlib and Seaborn. So Mathplotlib is a basic data visualisation library that is there in Python and which we have used for building visualisations like comparative charts, line graphs and also trying to fit a extrapolated curve so on and so forth.

Now Seaborn is a data visualisation library which is built on top of Mathplotlib and its main purpose is to provide a high-level interface for building attractive and informative graphics. The main advantage of having Seaborn is that your graphics become more aesthetically pleasing and are well structured. Now this will connect us back to the theoretical elements that we have studied in the course where you have to think about the accessibility, affordability and acceptance of the visuals that you are building for your audience.

In this aspect the affordances and the aesthetics can be well built into the visualisation using the features that are there in the Seaborn library and Seaborn library also has some kind of what I would call an intuitive functioning where the numerical averages are the measures that you would generally like to present in a graphic or automatically computed whereas in Mathplotlib you have to parse those commands specifically to get those outputs. We will see in the examples how Seaborn is a significant improvement over Mathplotlib and for presenting graphics. However having said this Mathplotlib still becomes the main core data visualisation library that you would like to use for doing your data visualisation works while exploring the data and may use Seaborn for presenting it on the dashboards or in the reports.

So there is good utility for both the libraries but however Seaborn is more sophisticated and produces more elegant graphs. Now without further delay let us understand what are the key features of Seaborn. So it is seamlessly integrated with Pandas data frames.

This is one of the key attributes of Seaborn as compared to Mathplotlib. In Mathplotlib you have to specify the columns and then the interaction with the existing data set is much more I would care rough or it is more a staggered whereas in case of Seaborn it is more seamless and we will see how does it make a difference while calling out the functions for building the graphics. Now there is a specialised support for categorical variables and to as I was mentioning when you are looking at different categories of variables and there is large continuous time series or the large cross-section of data you would like to look at the average statistics obviously and then in a Mathplotlib environment you have to pass the mean and average commands where in Seaborn it would automatically be computed and showcased and then functions and options are there to visualise linear regression models automatically and beautiful default colour palettes and visual styles add to the beauty of visualisations that are produced by Seaborn.

So now let us take up the setup of importing Seaborn into the work environment and from there we will see how it works as compared to the Mathplotlib. So I'm opening a new notebook to simultaneously run the command so that you will understand how it works. So what are we are trying to do is we are trying to construct an exam score based on the study hours trying to relate the exam score to the study hours and there are three courses right so this is what we are trying to create this is the data frame that we are trying to create.

So as usual first we will import the necessary libraries into our work environment we need Mathplotlib because we are trying to compare then Seaborn as SNS this is a standard thing you can import and then Pandas and NumPy so you are importing all the four required libraries into your work environment. So once again just as a reminder you can use this Gemini feature within the Colab environment to ask your doubts if you are stuck with a code error or you want to try something you can always type in there and ask for the help. So I'm just creating a sample data frame I am just copying this and as I just explained it is nothing but trying to relate the study hours to the exam scores okay random we are just giving random study hours to 50 students and then we are trying to generate exam scores for 50 students right and then across three subjects so if you see you will have the study hours then the exam score for the three courses.

So I'm just running the selection it is done now let us look at the df.head to understand how does our data frame look like this is our data frame looks like you have the study hour for student one then exam score exam score is given by this particular formula where it is three times then it is the courses mathematics then history then science so this is how you are getting it for the students right or 50 entries so 50 I think I get the entire space yeah so I get the entire space and in fact the subject choice is also random it is not all three subjects it is the course is math science one of the math science and history so it generates 50 times by picking choosing random subjects math history and science so this is the structure of our data so now let us get to the step of visualising the data so now let us see how syntax of MATLAB versus seaborn varies as I was mentioning the matlab matplotlib library you are more verbose and most of time require you extracting exact columns manually whereas seaborn it takes data frame and column names as direct arguments and makes the coding and makes the interaction more intuitive and easy so let us try to plot this scatterplot using the matplotlib commands that we have seen earlier so we are just doing plot dot scatter and what are we do we are subsetting the data frame for study hours versus studies exam score and then the title is matplotlib and we just want to see how does it look like so you can see as the study hours increases your exam score is likely to increase given the linear function the way we have defined and you are having the scatter plot now look at the same thing what is happening in seaborn so in seaborn you are just trying to do the same thing right so but look at the be I'll just take one more code cell yeah so I'm just look at the core so in seaborn look at the difference sorry I'll just paste it here so that you can compare these two yeah so look at the way the code is written SNS dot scatter plot PLT dot scatter both are similar there is nothing much of a difference but here you have to specify the X element and Y element right so yes what are you doing for X you are defining X is equal to DF of the particular column in the data frame DF you take the exact X variable to be study hours and Y variable to be the exam score this is how you are plotting in the matplotlib whereas if you come to the seaborn architecture or seaborn syntax this simply you are parsing the data frame data is coming from DF and X is simply study hours and Y is equal to exam score the seaborn is studying is taking the entire data frame and you can just call the specific columns X and Y instead of subsetting them computationally it is like subsetting the this data frame and then again and then using the scatter plot here the entire data frame is taken then you are plotting the two columns then you have the seaborn title then we are trying to show you get the same result but the way it is achieved is different and you can also see the bubble size being lesser in seaborn right and the automatically the titles have come on the x-axis as study hours and y-axis as exam score where is in matplotlib you had to give X label and Y label here in seaborn it automatically gave the X and Y label saving time and also immediately giving you while even though you are doing the exploration work giving an understanding what is there on the x-axis and what is there on the y-axis now let us see go further and explore the differences now let us understand the difference between seaborn and matplotlib in terms of the aesthetics okay so we are just again using the same exam score data but here we will be trying to build a histogram and see how aesthetically it looks different remember your graphics should be aesthetically pleasing so that you can get your audience attention much more in an efficient manner and help them understand the content that you are getting across we all prefer something that is a stick is typically presented as compared to something that is given without much of a importance to how it is designed or presented to take the best example is garnished hood versus not garnished food now coming to the code the code is almost similar we are not working on the code now we understood this we are subsetting a column in the matplotlib environment here we pass the data into data frame mean and then take the particular column that we want to show and then taking the code for running so I'm just taking this and just running it so you can see the default histogram of matplotlib which is more it is nice but it is striking in terms of it is very gaudy at the edges and the edges are very bright but as compared to seaborn is more smoother and pleasing on the edge while conveying the same information but in a more elegant fashion and this is one of the key differences between matplotlib and seaborn where the elegance and the aesthetics are built into the fun basic graphics itself so that they look naturally very pleasing to the audience are the persons who are looking at the data right so this is one more interesting feature of seaborn and again as in last case it has the exam score which is the x-axis and count as the y axis so you need not worry about what are the x and y levels you did not specify them it has automatically taken the exam score okay yeah sorry so this is what you have done here right so now let us take one more example the statistical plots in matplotlib versus seaborn what are statistical plots statistical plots are the ones which present the statistical in the central tendencies are the measures that you want to know about the underlying data so that you can form some conclusions about them right so the most commonly used statistical measure is average you have median dispersion standard deviation right all these things come into are very handy in understanding the underlying data so let me pause a second for here and get back you to the seven stages of data visualisation and the key element of your data visualisation journey is data exploration where you would be needing to know the averages and all then form some opinion about how to present the data as a visual if the averages are very distant and the standard deviation is high you might choose a different kind of a plot or you plot in two different ranges to show the distribution within the data so while presenting such visualisations giving getting averages automatically without having to parse them as commands is very useful now let us see how you do you calculate statistics in a matplotlib environment versus what you do in a seaborn environment right so I'm just taking the code here then I think it here so let us see so what you want to have the mean score average score for this as per the subject so first you have to create a subset right based on the exam score you take the exam score and then subset it by course right you have to use DF dot group by function you are grouping by course and what are you grouping you are grouping exam score and then you are calculating the mean of that particular thing so how does it look like so if you go back to the data frame you are first collecting all the math course then summing up and then you are taking the subset of exam scores for math and then averaging it this is how the logic is working so and that is what you have parsed into the command here for matplotlib you take the DF and then you group it by the course maths science and history and then you have the exam score column within that subset and do the mean of that particular column then you are trying to plot a bar X axis is course mean dot index so you have the course mean index then height is the course mean values so course means means you will get the course we have defined the course means so you'll have three course means because you will we are grouping by three courses so those will become the index on X axis which is maths science and history and then the height is of course the mean values again this is coming from course means so the course means is the variable that you have just created or the array that you have created by a subsetting the and doing the average of the elements from the data frame then you are saying it is matplotlib manual average because you have to write all these codes and then you are trying to so this is how I got the average history math science then you have the average values here then it is on the X axis matplotlib annual manual average now in SNS I need not do all this I can just simply pass the data frame into the data then I say X is equal to course and Y is equal to exam score that's it and nothing else I need to do even this is plot title and show is not required let me just show you this part run selection so I am just getting this so it's very interesting to see the way it is done so it not only computed the averages of the course without asking me asking to compute the average and then it has titled it course exam score and most importantly it followed the just like principles and there were and it found out the averages are decreasing across subjects and ordered it in a particular fashion as compared to the matplotlib plot so in matplotlib we had history math and science though math was higher than history so visually if you remember the main purpose when you look at the graph what you do you detect the assembly then try to do the comparisons the matplotlib graph is not enabling comparisons in an efficient manner because yes you have identified the job geographical object as the vertical bar representing the average score but then at that particular level you are not able to easily compare whether average score in math is higher than history than size by and then by how much so these comparisons are not coming effectively and the principles of visualisations require that if there is a natural order or if your data can be ordered in a certain fashion please do present it in that format so that the audience can easily grab the underlying trend so the math average is higher than history than in science or history or science in fact looking at the matplotlib graph you might feel that history is little higher than science but it is very close so once you look at the seaborn graph it is very clear that history and science averages are almost close and math is definitely different and the bars that you see at the top of each of these graphs are vertical bars are confidence intervals which shows the range with which the average can vary in that particular course given the sample size so these statistical properties are well presented in seaborn as compared to matplotlib now let us go further yeah so we can see an interactive example for combining the matplotlib and seaborn okay yeah so now let us try to see how one can integrate both matplotlib and seaborn to the same visualisation and use it for building something interactive and custom customised so we look at this example to see seaborn to create a plot and then use matplotlib to customise it okay so yeah you will also need that are commented out let us see okay so first what we are doing we are using the same data set that we have created till now right so we just have the data and we are trying to plot the box plot using the seaborn SNS SNS dot box plot same the data frame is data is passed into the data frame then you have the x-axis as course so you have three different courses and you want y to see the distribution of exam score across the individuals so what does a box plot to box plot is simply nothing but it shows the distribution mainly focussing on how widely or it is how widely it is distributed or how compactly the distribution is it presents the median the 25th percentile 75th percentile and the interquartile range and outliers anything beyond that so now we just got this math history and science again this is ordered in terms of the median values where in math the median value is higher compared to history than in science but the dispersion is more in history between the 25th and 75th percentile values then you have the first quartile values then you have science where the median is low the people who are there are many more people who are above the value a median value the distribution is little skewed than compared to math right so you have this characteristics coming from the box plot then we are trying to customise this using the math plot lip function first we are trying to add title so uncomment the line and then add the custom line so I'm just taking it here okay I just want to plot title custom title added with math plot clip so PLD title then you are doing the Y label as final score you are adding the exam score instead of exam score you are adding the Y label as final score that has now changed as final score right and then you add a horizontal line representing the overall average score okay so that you will try to see how the average score is across so look at this particular code your PLT axis H line okay axis H line is a horizontal line and it is coming horizontally and how are you getting it so you need to do it based on the exam score you want to want average so you are using the function mean colour is red so then you have line style as this so the double and then so this is the average score so and you are able to see that history is representing the average score across all this and then maths as above average science as below average which is the essence that you got from the bar graph but this is more precisely presented maths is much math average in maths is much higher than the average in science which is below the overall average so the distance of maths average is much higher than from the overall average as compared to the fall of science average from the overall average so this is interesting you just plotted using the seaborn command but you also integrated the features from the math plot now let us take this a step further and try to understand how to how we can use continuity oh sorry before we get into this continuity let us see the summary table and see a quick comparison between math problems in seaborn math part of this features levels are low level it is very high in seaborn high customisation but here the object is statistical visualisation for seaborn you can customise a lot with the math thought loop which is the strength and weakness also in some sense because you would be requiring to pass each and every command again and again whereas seaborn it is more of understanding underlying distributions easily using the exact statistical properties now more webOS which is concise in syntax is very intuitive data input is primary arrays and lists that so you have to subset and give them seaborn the inputs are optimised for pandas data frame so they can directly passed as data the data is passed as data frame then you can directly call the columns and you have a stably pleasing defaults in seaborn as compared to basic features in math.lib statistics you need manual calculation like in you have to write the mean okay or median every time then it is built in in seaborn so these are the basic differences between these two.

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